

Backend Onboarding Document

Onboarding Documents

New backend members should begin by reading the following documents included in the project (in 'backend/docs' from GitHub repo). These provide essential context for authentication, ingestion, security, and frontend–backend integration:

Authentication Document

Explains the backend authentication system, including:

- JWT authentication
- Bcrypt password hashing
- Role-Based Access Control (RBAC)
- Layered architecture (routes → controller → service → repository)
- Login, registration, token generation, and protected routes
- Security features such as account locking and failed login tracking

This document helps new members understand how user authentication works and how protected routes are enforced.

Database Implementation Documentation

Explains how the backend stores and retrieves IoT sensor data using PostgreSQL.

Includes:

- Controller → Service → Repository architecture
- Unified wide-format ingestion for CSV and ThingSpeak
- Automatic dataset creation
- Dataset and series endpoints
- Analysis endpoints
- Ingestion endpoints (CSV + ThingSpeak)

This document helps new members understand how data flows from ingestion → database → API responses.

Contract With Frontend

Defines how the backend and frontend should communicate for the next sprint, including:

- API endpoint usage
- Request/response formats
- Error handling conventions
- Integration expectations
- Responsibilities of backend vs frontend
- Recommendations for replacing mock data with real API calls

This ensures the frontend team can reliably connect to backend APIs without guesswork.

Threat Modelling Document

Provides a STRIDE-based analysis of security risks across:

- IoT sensors and edge devices
- Communication networks
- APIs and backend services
- Databases and cloud infrastructure
- Authentication and access control
- Monitoring and logging

This document helps backend members understand potential vulnerabilities and required mitigation strategies.

How to Set Up the Development Environment

Requirements

- Node.js 18+
- PostgreSQL installed locally
- pgAdmin (optional)
- VS Code

Setup Steps

1. Clone the repository.
2. Install backend dependencies: `npm install`
3. Create a `.env` file if doesn't exist like this

```
#Express server port, uncomment and add yours, otherwise it will default to 3000 in server.js
#PORT=3000s

#Load path for dataset here
PROCESSED_DATA_PATH=./mock_data/processed_data.json

#for pool.js
DB_USER=postgres
DB_PASSWORD=postgrespw
DB_HOST=localhost
DB_PORT=5432
DB_NAME=IoTDatabase
```

4. Initialise the database schema: `psql -f schema.sql`
5. Start the backend server: `npm start`

Optional: Test Ingestion Tools

- Ingest ThingSpeak data: `node src/dataIngestion/thingSpeakInjest.js <channelId>`
- Preview a ThingSpeak channel: `node src/dataIngestion/previewThingSpeak.js <channelId>`

Accessing Resources

- GitHub repository: [DataBytes-Organisation/Intelligent-IoT-Data-Management](https://github.com/DataBytes-Organisation/Intelligent-IoT-Data-Management)

- High level MS planner for our entire IoT team: [Intelligent IoT Data Management](#)
- Detailed MS planner for backend team (can be updated by new leads): [Backend Team Tasks Detailed](#)

If links are broken, request updated access in Teams!

Full Access

- Backend Leader
- Junior Backend Leader
- Backend Members
- Backend Coordinator

Limited Access

- Frontend team (API only)
- Data Science team (datasets + series endpoints only)

Team Member Profile Template for students next trimester

Each backend member should provide:

- Name
- Role (Leader, Junior Leader, Coordinator, Member)
- Technical strengths
- Preferred tasks
- Availability
- Preferred grade target
- Experience with Node/PostgreSQL
- Comfort level with CSR architecture

Position Descriptions & Preferred Grade Targets

Senior Backend Leader

- Owns architecture, schema, ingestion, and code quality
- Reviews all PRs
- Coordinates with frontend and data science
- Implements assigned tasks
- Preferred grade target: HD

Junior Backend Leader

- Supports PR reviews
- Helps onboard new members
- Assists with architecture decisions
- Implements assigned tasks

- Preferred grade target: D-HD

Backend Coordinator (For Senior Leads & Backend Leads)

- Implements assigned tasks
- Maintains documentation
- Tracks tasks and deadlines
- Ensures team alignment
- Preferred grade target: C-D

Backend Member

- Implements assigned tasks
- Follows CSR architecture
- Writes clean, testable code
- Preferred grade target: P-C